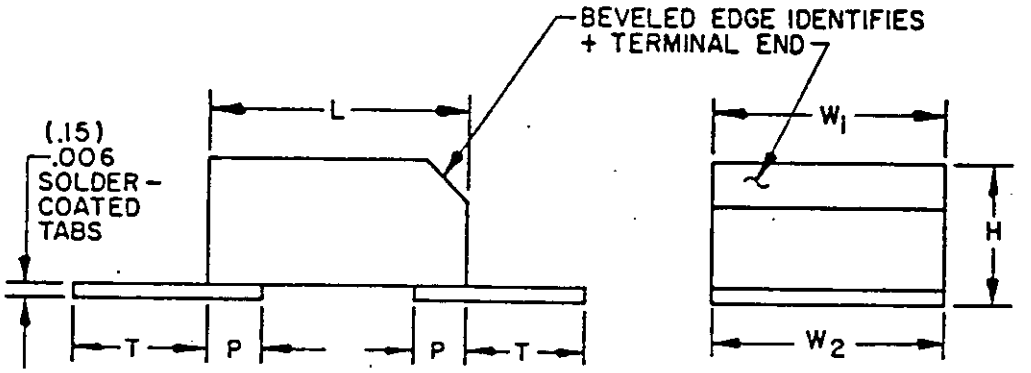
MILITARY SPECIFICATION SHEET

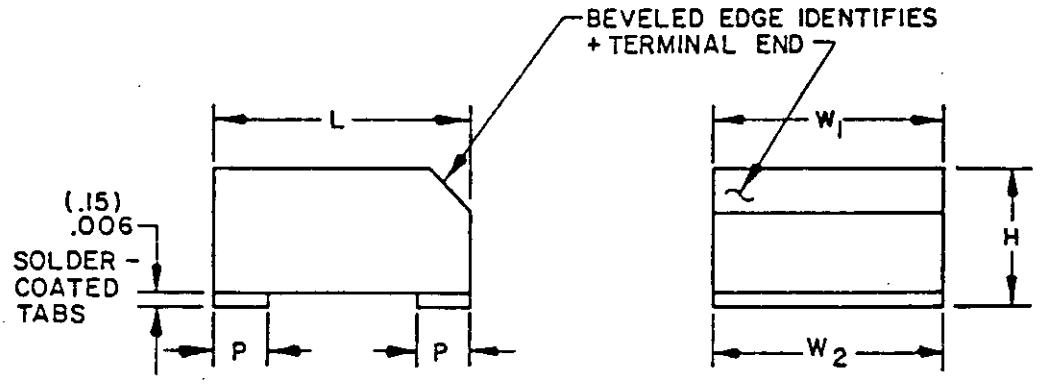
CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY, STYLES CWR03 AND CWR04

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-55365.



STYLE CWR03 (ENCAPSULATED)



STYLE CWR04 (ENCAPSULATED)

FIGURE 1. Styles CWR03 and CWR04 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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Case size	L	(Case) W_1	(Tabs) W_2	H	P	T
	±.005 (0.13)	±.010 (0.25)	+.010 (0.25) -.005 (0.13)	±.005 (0.13)	±.010 (0.25)	±.010 (0.25)
A	.180 (4.57)	.100 (2.54)	.090 (2.29)	.070 (1.78)	.030 (0.76)	.150 (3.81)
B	.180 (4.57)	.100 (2.54)	.090 (2.29)	.100 (2.54)	.030 (0.76)	.150 (3.81)
C	.320 (8.13)	.180 (4.57)	.170 (4.32)	.070 (1.78)	.050 (1.27)	.110 (2.79)
D	.320 (8.13)	.180 (4.57)	.170 (4.32)	.100 (2.54)	.050 (1.27)	.110 (2.79)
E	.320 (8.13)	.180 (4.57)	.170 (4.32)	.195 (4.95)	.050 (1.27)	.110 (2.79)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. These capacitors are designed for mounting by dip soldering, welding, reflow soldering, or other conventional means.

FIGURE 1. Styles CWR03 and CWR04 capacitors - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Chip.

Material: Tantalum.

Leads: Tab.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-C-55365).

Operating temperature range: -55°C to +125°C.

Failure rate level: B (0.1 percent), C (0.01 percent), D (0.001 percent), E (0.0001 percent), M (1.0 percent), P (0.1 percent), R (0.01 percent), and S (0.001 percent).

DC leakage (DCL): See table I.

Capacitance (Cap.): See table I.

Cap. tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition E.

Stability at low and high temperatures: In accordance with MIL-C-55365.

Step 1 (+25°C):	DCL	-	See table I.
	Cap.	-	Shall not exceed tolerance of table I value.
	DF	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Shall not exceed ±10 percent of initial measurement.
	DF	-	See table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.
Step 4 (+85°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±10 percent of initial +25°C measurement.
	DF	-	See table I.
Step 5 (+125°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±15 percent of initial +25°C measurement.
	DF	-	See table I.
Step 6 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.

Surge voltage: In accordance with MIL-C-55365.

DCL	-	See table I.
ΔCap.	-	Within ±5 percent of initial value.
DF	-	See table I.

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Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C: DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial measured value.
DF - See table I.

At 85°C: DCL - See table I.

At 125°C: DCL - See table I.

10,000 hours:

At 25°C: DCL - 200 percent of initial requirement.
ΔCap. - Shall not exceed ±10 percent of initial measured value.
DF - See table I.

At 85°C: DCL - 200 percent of initial requirement.

At 125°C: DCL - 200 percent of initial requirement.

TABLE I. Styles CWR03 and CWR04 characteristics.

Type designation 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	
	Volts	μF	μA	μA	μA	%	%	%	
CWR0-B-185--	3	1.8	1.0	10	15	4	6	8	A
CWR0-B-225--	3	2.2	1.0	10	15	4	6	8	A
CWR0-B-395--	3	3.9	1.0	10	15	4	6	8	B
CWR0-B-475--	3	4.7	1.0	10	15	4	6	8	B
CWR0-B-126--	3	12.0	1.0	10	15	6	8	10	C
CWR0-B-156--	3	15.0	1.0	10	15	6	8	10	C
CWR0-B-396--	3	39.0	2.3	23	35	6	8	10	D
CWR0-B-476--	3	47.0	2.8	28	42	6	8	10	D
CWR0-B-826--	3	82.0	4.9	49	74	8	10	12	E
CWR0-B-107--	3	100.0	6.0	60	90	8	10	12	E
CWR0-D-125--	6	1.2	1.0	10	15	4	6	8	A
CWR0-D-155--	6	1.5	1.0	10	15	4	6	8	A
CWR0-D-275--	6	2.7	1.0	10	15	4	6	8	B
CWR0-D-335--	6	3.3	1.0	10	15	4	6	8	B
CWR0-D-825--	6	8.2	1.0	10	15	6	8	10	C
CWR0-D-106--	6	10.0	1.2	12	18	6	8	10	C
CWR0-D-276--	6	27.0	3.2	32	48	6	8	10	D
CWR0-D-336--	6	33.0	4.0	40	60	6	8	10	D
CWR0-D-566--	6	56.0	6.7	67	101	8	10	12	E
CWR0-D-686--	6	68.0	8.2	82	123	8	10	12	E
CWR0-F-824--	10	0.82	1.0	10	15	4	6	8	A
CWR0-F-105--	10	1.0	1.0	10	15	4	6	8	A
CWR0-F-155--	10	1.5	1.0	10	15	4	6	8	B
CWR0-F-185--	10	1.8	1.0	10	15	4	6	8	B
CWR0-F-225--	10	2.2	1.0	10	15	4	6	8	B
CWR0-F-565--	10	5.6	1.1	11	17	4	6	8	C

See footnote at end of table.

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TABLE I. Styles CWR03 and CWR04 characteristics - Continued.

Type designation 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	
	Volts	µF	µA	µA	µA	%	%	%	
CWR0-F-685--	10	6.8	1.4	14	21	6	8	10	C
CWR0-F-126--	10	12.0	2.4	24	36	6	8	10	D
CWR0-F-156--	10	15.0	3.0	30	45	6	8	10	D
CWR0-F-186--	10	18.0	3.6	36	54	6	8	10	D
CWR0-F-226--	10	22.0	4.4	44	66	6	8	10	D
CWR0-F-336--	10	33.0	6.6	66	99	6	8	10	E
CWR0-F-396--	10	39.0	7.8	78	117	6	8	10	E
CWR0-F-476--	10	47.0	9.4	94	141	6	8	10	E
CWR0-H-564--	15	0.56	1.0	10	15	4	6	8	A
CWR0-H-684--	15	0.68	1.0	10	15	4	6	8	A
CWR0-H-125--	15	1.2	1.0	10	15	4	6	8	B
CWR0-H-395--	15	3.9	1.2	12	18	4	6	8	C
CWR0-H-475--	15	4.7	1.4	14	21	4	6	8	C
CWR0-H-825--	15	8.2	2.5	25	38	6	8	10	D
CWR0-H-106--	15	10.0	3.0	30	45	6	8	10	D
CWR0-H-276--	15	27.0	8.1	81	122	6	8	10	E
CWR0-J-274--	20	0.27	1.0	10	15	4	6	8	A
CWR0-J-334--	20	0.33	1.0	10	15	4	6	8	A
CWR0-J-394--	20	0.39	1.0	10	15	4	6	8	A
CWR0-J-474--	20	0.47	1.0	10	15	4	6	8	A
CWR0-J-564--	20	0.56	1.0	10	15	4	6	8	B
CWR0-J-684--	20	0.68	1.0	10	15	4	6	8	B
CWR0-J-824--	20	0.82	1.0	10	15	4	6	8	B
CWR0-J-105--	20	1.0	1.0	10	15	4	6	8	B
CWR0-J-185--	20	1.8	1.0	10	15	4	6	8	C
CWR0-J-225--	20	2.2	1.0	10	15	4	6	8	C
CWR0-J-275--	20	2.7	1.1	11	17	4	6	8	C
CWR0-J-335--	20	3.3	1.3	13	20	4	6	8	C
CWR0-J-475--	20	4.7	1.9	19	29	4	6	8	D
CWR0-J-565--	20	5.6	2.2	22	33	4	6	8	D
CWR0-J-685--	20	6.8	2.7	27	41	6	8	10	D
CWR0-J-126--	20	12.0	4.8	48	72	6	8	10	E
CWR0-J-156--	20	15.0	6.0	60	90	6	8	10	E
CWR0-J-186--	20	18.0	7.2	72	108	6	8	10	E
CWR0-J-226--	20	22.0	8.8	88	132	6	8	10	E
CWR0-M-104--	35	0.10	1.0	10	15	4	6	8	A
CWR0-M-124--	35	0.12	1.0	10	15	4	6	8	A
CWR0-M-154--	35	0.15	1.0	10	15	4	6	8	A
CWR0-M-184--	35	0.18	1.0	10	15	4	6	8	A
CWR0-M-224--	35	0.22	1.0	10	15	4	6	8	A
CWR0-M-274--	35	0.27	1.0	10	15	4	6	8	B
CWR0-M-334--	35	0.33	1.0	10	15	4	6	8	B
CWR0-M-394--	35	0.39	1.0	10	15	4	6	8	B
CWR0-M-474--	35	0.47	1.0	10	15	4	6	8	B
CWR0-M-824--	35	0.82	1.0	10	15	4	6	8	C
CWR0-M-105--	35	1.0	1.0	10	15	4	6	8	C
CWR0-M-125--	35	1.2	1.0	10	15	4	6	8	C
CWR0-M-155--	35	1.5	1.1	11	17	4	6	8	C
CWR0-M-185--	35	1.8	1.3	13	20	4	6	8	D
CWR0-M-225--	35	2.2	1.5	15	23	4	6	8	D
CWR0-M-275--	35	2.7	1.9	19	29	4	6	8	D
CWR0-M-335--	35	3.3	2.3	23	35	4	6	8	D

See footnote at end of table.

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TABLE I. Styles CWR03 and CWR04 characteristics - Continued.

Type designation 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	
	Volts	µF	µA	µA	µA	%	%	%	
CWR0-M-395--	35	3.9	2.7	27	41	4	6	8	D
CWR0-M-475--	35	4.7	3.3	33	50	4	6	8	E
CWR0-M-565--	35	5.6	3.9	39	59	4	6	8	E
CWR0-M-685--	35	6.8	4.8	48	72	6	8	10	E
CWR0-M-825--	35	8.2	5.7	57	86	6	8	10	E
CWR0-M-106--	35	10.0	7.0	70	105	6	8	10	E
CWR0-N-683--	50	0.068	1.0	10	15	4	6	8	A
CWR0-N-823--	50	0.082	1.0	10	15	4	6	8	A
CWR0-N-104--	50	0.10	1.0	10	15	4	6	8	B
CWR0-N-124--	50	0.12	1.0	10	15	4	6	8	B
CWR0-N-154--	50	0.15	1.0	10	15	4	6	8	B
CWR0-N-184--	50	0.18	1.0	10	15	4	6	8	B
CWR0-N-224--	50	0.22	1.0	10	15	4	6	8	C
CWR0-N-274--	50	0.27	1.0	10	15	4	6	8	C
CWR0-N-334--	50	0.33	1.0	10	15	4	6	8	C
CWR0-N-394--	50	0.39	1.0	10	15	4	6	8	C
CWR0-N-474--	50	0.47	1.0	10	15	4	6	8	C
CWR0-N-564--	50	0.56	1.0	10	15	4	6	8	C
CWR0-N-684--	50	0.68	1.0	10	15	4	6	8	C
CWR0-N-824--	50	0.82	1.0	10	15	4	6	8	D
CWR0-N-105--	50	1.0	1.0	10	15	4	6	8	D
CWR0-N-125--	50	1.2	1.2	12	18	4	6	8	D
CWR0-N-155--	50	1.5	1.5	15	23	4	6	8	D
CWR0-N-185--	50	1.8	1.8	18	27	4	6	8	E
CWR0-N-225--	50	2.2	2.2	22	33	4	6	8	E
CWR0-N-275--	50	2.7	2.7	27	41	4	6	8	E
CWR0-N-335--	50	3.3	3.3	33	50	4	6	8	E

1/ Complete type designation shall include additional symbols to indicate style, termination finish, capacitance tolerance, and FR level.

Marking: Capacitors shall be marked in accordance with method I of MIL-STD-1285. The capacitance, voltage, and "J" brand shall be indicated on the capacitors.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 11, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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